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EDUCATION

DePaul University - MS in Data Science (GPA 4/4) Chicago, IL USA |Mar 2025 - Present
Coursework: Machine Learning, Programming ML, Advanced Modeling and Analysis, Mining Big Data

FAST-NUCES - MS in Data Science (Gold medal |GPA 3.94/4) Karachi, PK | Jun 2020 - Aug 2022
Coursework: Machine Learning, Deep Learning, Natural Language Processing, Applied Computer Vision

Mohammad Ali Jinnah University - BS in Computer Science (GPA 3.74/4) Karachi, PK | Jan 2013 - Dec 2016
Coursework: Computer Programming, OOP, Data Structure, Design and Analysis of Algorithm, Database systems, AI

EXPERIENCE

DePaul University - NIH R15 VS-EDGE Research Assistant Chicago, IL, USA | May 2026 - Present

- Developing AI-based computer vision systems for medical diagnosis as part of the NIH-funded VS-EDGE project in collaboration with University of Chicago.
- Implementing and evaluating machine learning and deep learning models for medical imaging research, including the development of visualization tools, explainable AI interfaces, and evaluation frameworks to support CAD systems and clinical decision-making workflows.

DePaul University - Graduate Research Assistant Chicago, IL, USA | Jun 2025 - Present

- Developing atlas-based and multimodal AI methods that combine radiomics, clinical, and deformation-based features for cardiac disease analysis.
- Developing a spatiotemporal deep learning framework for full-cycle cine MRI segmentation, building on benchmarking of U-Net and TransUNet architectures and loss functions.
- Analyzing gene expression data and applying machine learning methods to predict gene-related outcomes.

DePaul University - Research Assistant Chicago, IL, USA | Aug 2025 - Apr 2026

- Developed deep learning models using convolutional neural networks (CNNs) to automate foot posture measurement and classification, enhancing accuracy and efficiency in the Foot Posture Index Automation Project.
- Conducted image preprocessing, augmentation, and model optimization, while documenting research outcomes for dissemination in academic and clinical settings.

Aga Khan Education Services - Specialist, Software Solutions and Innovation Karachi, PK | Jun 2022 - Aug 2023

- Developed LSTM predictive models for student performance and enrollment forecasting using Python libraries such as Scikit-learn and TensorFlow. These models contributed to decision-making processes and increased efficiency by streamlining resource allocation.
- Conducted exploratory data analysis on examination datasets using Python, with libraries such as Pandas, NumPy and Matplotlib to identify trends and patterns. The actionable insights derived improved curriculum and exam development efficiency.

Aga Khan University - Simulation Technical Analyst, CIME Karachi, PK | Jun 2021 - May 2022

- Analyzed medical simulation data using Python, Power BI and Tableau to assess the efficacy of training programs, resulting in enhanced efficiency by pinpointing areas for improvement and optimizing decision-making processes.
- Engineered data pipelines with Python and SQL for seamless data ingestion, transformation, and storage, guaranteeing data integrity and accessibility for analysis, thereby improving operational efficiency.

Aga Khan Education Services - Senior IT Analyst Karachi, PK | Oct 2018 - May 2021

- Implemented innovative solutions to automate business processes and workflows, enhancing operational efficiency. Utilized programming languages like Python and Java to develop automation scripts and applications.
- Overhauled data management systems and reporting processes using SQL databases and Power BI, resulting in data accuracy improvement of 25%, and improving decision-making speed and capabilities throughout the organization.

TECHNICAL SKILLS

Languages & Databases: Python, R, C#, JavaScript, HTML, CSS, Bootstrap, SQL, MySQL, PostgreSQL, NoSQL, MongoDB
Tools & Technologies: Linux, Docker, Bash, AWS, Azure, GCP, Tableau, Power BI, MS Excel, Visual Studio, PyCharm, GitHub, PuTTY
Frameworks & Libraries: Pandas, NumPy, Scikit-learn, TensorFlow, Keras, PyTorch, OpenCV, Oracle APEX

PROJECTS & PUBLICATIONS

1. YOLOv5-M: A Deep Network for Medical Object Detection in Real-time
Modified YOLOv5 with Python and TensorFlow, trained and tested on various mask datasets, and used data augmentation through TensorFlow and Keras on Google Colab to expand the dataset size, achieving high accuracy and efficiency in medical imaging applications.

2. LSTM-based Model for Forecasting of COVID-19 Vaccines in Pakistan
Utilized Python, Pandas, and NumPy to preprocess and clean official datasets. Developed and trained an LSTM prediction model on Google Colab to accurately forecast COVID-19 vaccine distribution patterns.

3. Exploring the Impact of Preprocessing Techniques on Retinal Blood Vessel Segmentation Using a Study Group Learning Scheme
Implemented SGL model on 4 different datasets, used preprocessing techniques like CLAHE and MIN-MAX contrast stretching.

4. Understanding Tumor Micro Environment using Graph theory
Developed a framework based on Graph Theory, Gaussian Mixture Modeling and Fuzzy C-mean Clustering, for learning the cell characteristics in neoplastic lymphomas along with quantitative analysis.

5. Controlled Factorial Analysis of Architecture–Loss Interactions in Cardiac MRI Segmentation
Conducted a controlled factorial benchmarking study comparing U-Net and TransUNet across six loss configurations on the ACDC cardiac MRI dataset using 60 independent training runs with statistical validation.